

Advanced Computer Networks

Solar System Internet – Protocols, Mars Networking
and OPS-SAT Field-tests

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Content

1) Overview of Delay- and Disruption-tolerant Networking (DTN)

- Motivation for DTN protocols forming the fundament of interplanetary communication

2) IOAG report about Mars communication approaches

- Report focusses on an evolutionary approach for establishing a networking capabilities between Earth and Mars

3) Selected field-tests leveraging ESA's OPS-SAT

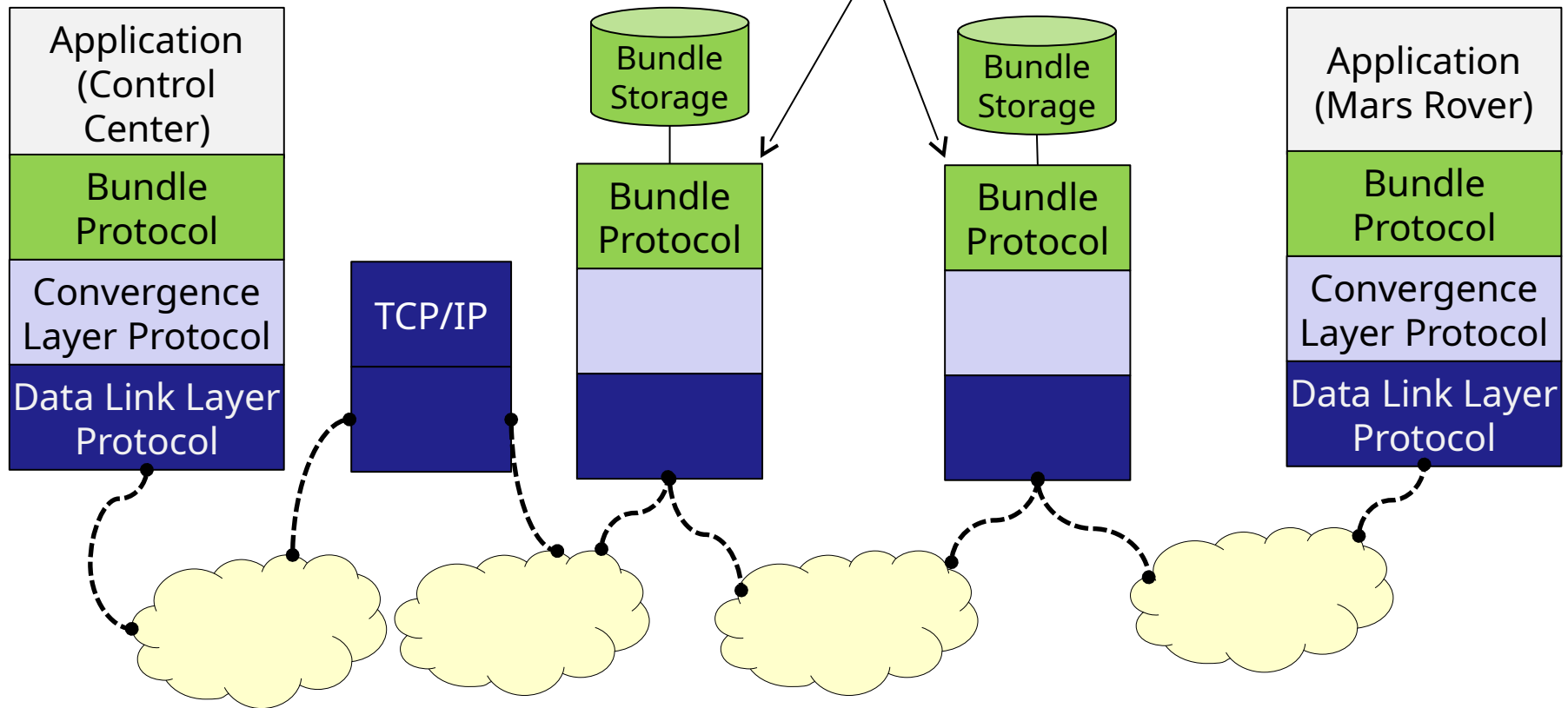
- Tests explore / demonstrate selected aspects for interplanetary communication

Challenges of Space Communication

- Long and variable transmission delay
 - e.g. round trip time (RTT) between Earth <-> Mars is between ~6 and ~40 minutes
 - Frequent and long connection interruptions
 - However: almost always plannable
 - High error rate
 - up to 10^{-3}
 - Limited energy supply
 - Asymmetric connections
 - up to 1000:1
- TCP/IP protocol stack not applicable

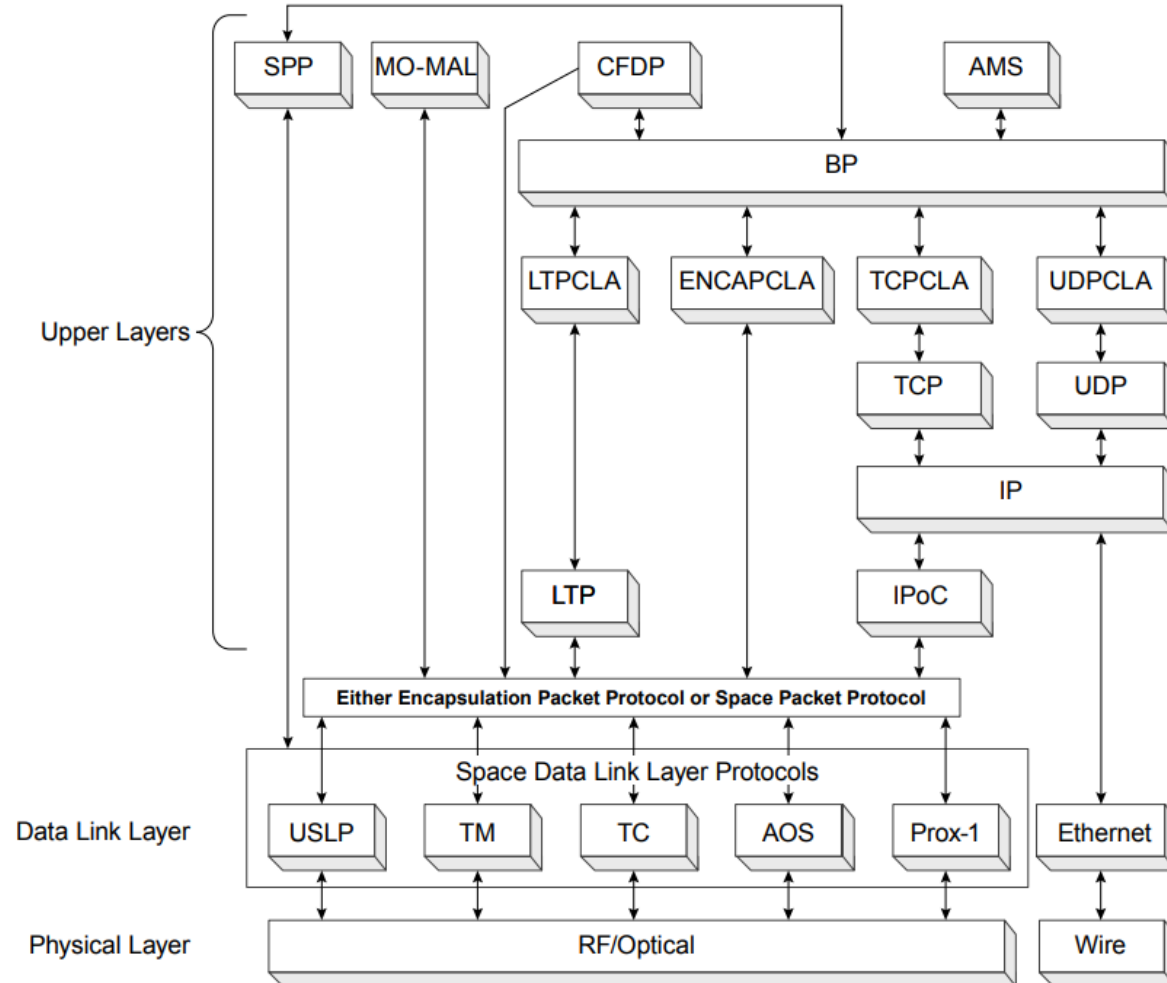
Overview of Bundle Protocol

Intermediary agents assume responsibility for the transfer ("custody") of a bundle (only in case of BPv6)



- Für Details der Architektur siehe RFC 4838

Space Networking Protocol Stack

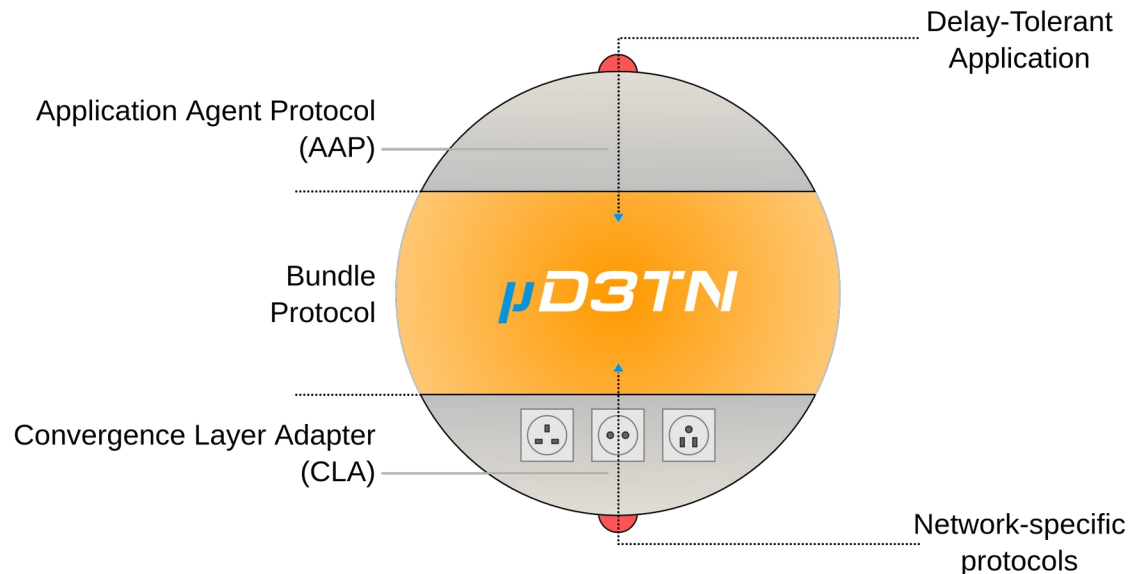


Taken from [1], Figure 2-1

Several Space Protocols Standardized by Consultative Committee for Space Data Systems (CCSDS)

μD3TN – Overview of a DTN protocol implementation

- μD3TN is an open-source DTN protocol implementation based on μPCN created in 2014
- First BPv7 implementation that has been created (mentioned under the name μPCN in BPv7 draft specification)
- Modular design offering several convergence layer adapters
- Simple integration of delay-tolerant applications due to Application Agent Protocol



Source code available via:
<https://gitlab.com/d3tn/ud3tn>

Background IOAG Report [2]

- Report is provided by the „Interagency Operations Advisory Group“ (IOAG)
- “IOAG was founded by the Interoperability Plenary (IOP) to:
 - 1) Understand issues related to interagency interoperability and other space communications matters;
 - 2) Identify solutions complying with IOP policies; and,
 - 3) Recommend resolutions to the IOP for specific actions by the IOP.” (From: <https://ioag.org/>)
- Report has been created by the Mars and Beyond Communications Architecture Working Group (MBC-AWG) with contributions from ASI, CNES, CNSA, ESA, ISRO, JAXA, NASA, and UK Space Agency

From: <https://ioag.org/>

IOAG MEMBERS



IOAG OBSERVERS



Structure of the Report

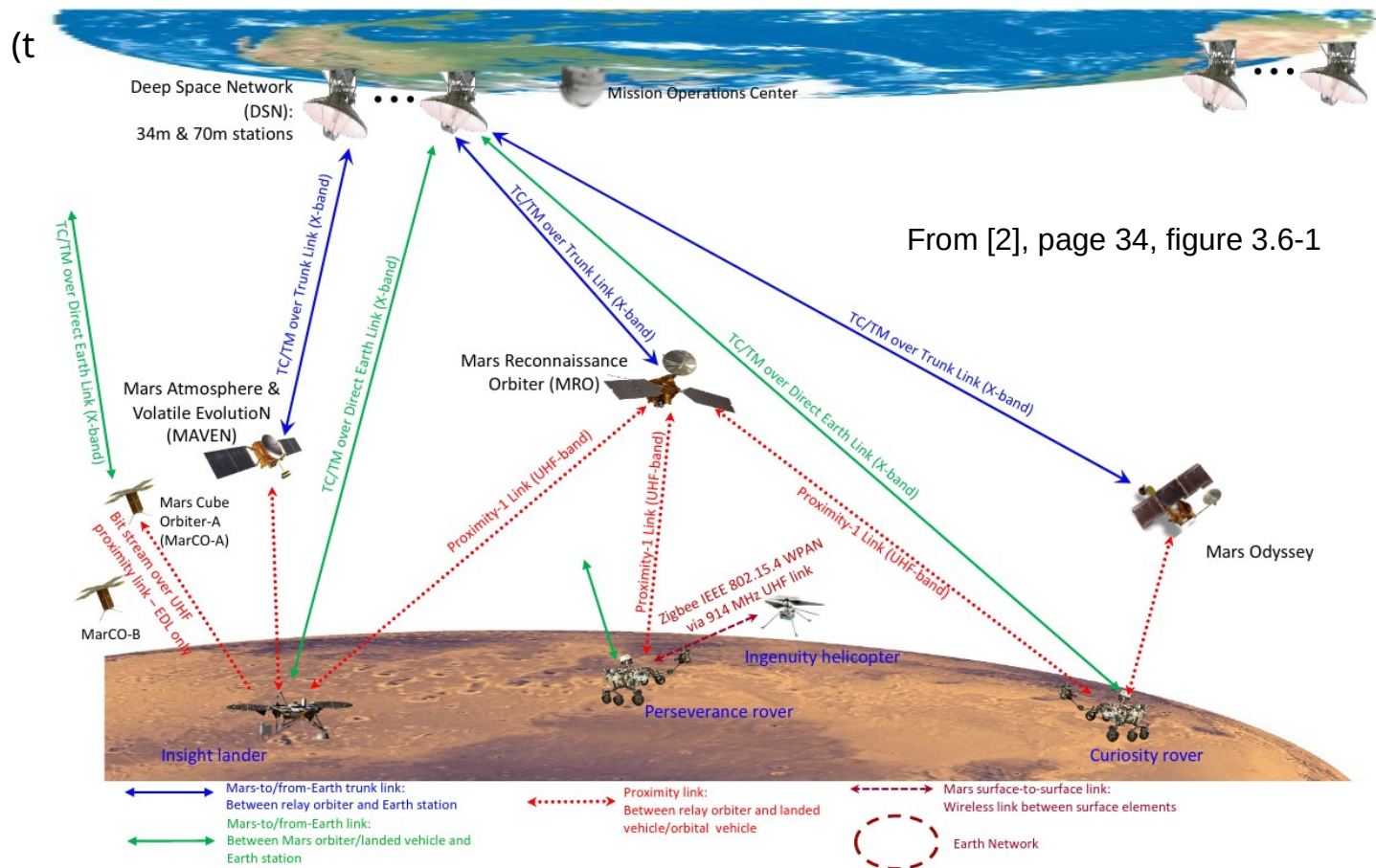
- The report consists of five parts:
 - 1) Introduction / motivation pointing out the high-relevance of a Mars network due to ~40 emerging missions with 60 assets on ground and in the Mars orbit in the next two decades
 - 2) Overview of Mars communication approaches applied and envisioned by ten different agencies (CNSA, ESA, ISRO, JAXA, ROSCOSMOS, NASA, UAE, UK Space Agency, ASI, CNES)
 - 3) Description and selection of frequency ranges, modulation, coding, ranging approaches, and link protocols with a clear focus on SFCG and CCSDS input and standards if possible
 - 4) Description of the Mars communication architecture including Mars relay, Mars surface and Earth networks plus the Mars network management architecture, navigation architecture and Mars network security architecture
 - 5) Conclusion and key recommendations for near-term, mid-term and long-term mission eras

Background Report

- Report took into account (see p. 6 f. of [2]):
 - The communication needs of all Mars missions planned by the member agencies for the next two decades
 - The architectural “requirements” as envisioned for the international Moon-to-Mars exploration initiative
 - The advancement of communications technology and the commercial providers of communication services
 - The CCSDS and industry standards
 - The ITU-R and SFCG frequency band allocations
 - The existing and planned new communication assets, capabilities, and services, at all space agencies and potential commercial providers, that are relevant to Mars communications

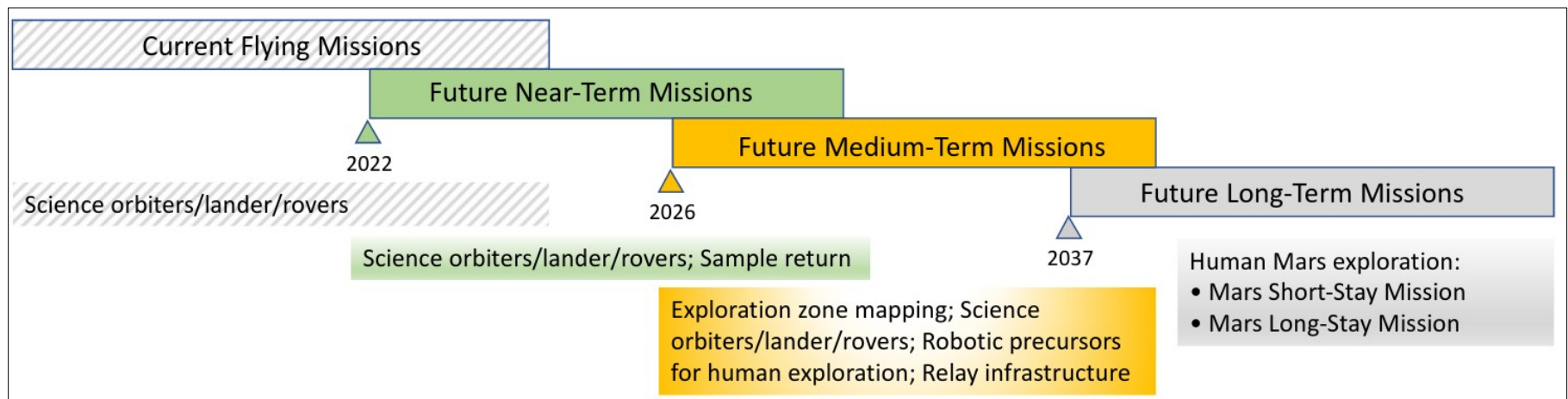
NASA's Mars Communication Capabilities

- Backbone of the end-to-end system is the direct-with-Earth (DWE) link using X-band and the proximity link based on UHF-band
- The UHF link for relay has been used by them for the entry, descent, and landing (EDL) and surface operations phases. During the EDL phase, through the UHF link with a science orbiter in view they were able to provide extensive information about the spacecraft critical events



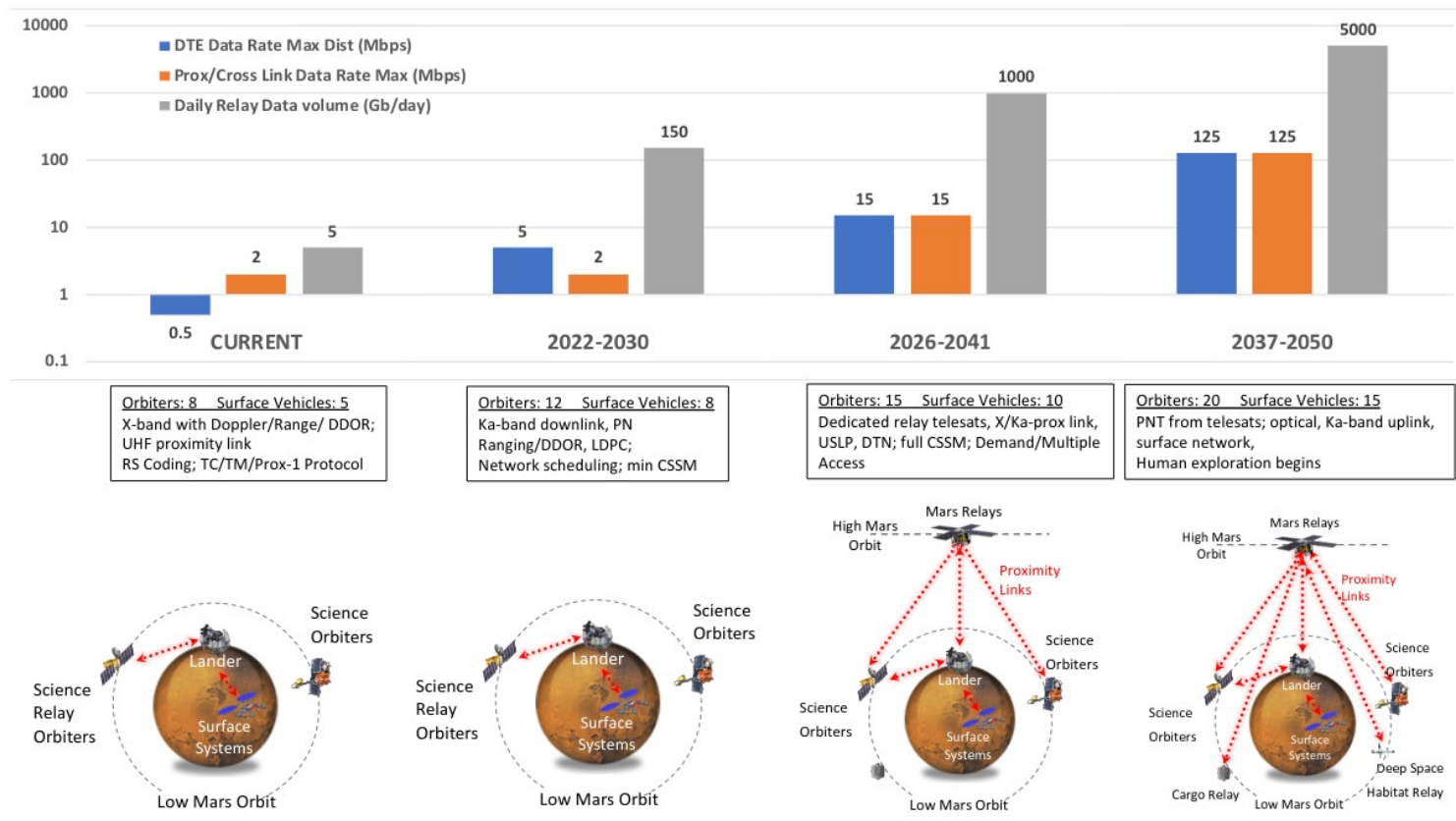
Current Missions and Mars Exploration Eras

- Current state:
 - Except two orbiters and the helicopter, all assets have the capability to communicate over the DWE and proximity link
 - All current missions rely on cross support by communication assets owned or operated by other agencies
 - However, several drawbacks, e. g.:
 - Limited relay capabilities (relaying is secondary function of science orbiters)
 - Constrained communication links mainly due to usage of X-band
 - Offered communication services are based purely on the capabilities enabled by the space data links and physical links (point-to-point links), no end-to-end networking features are available



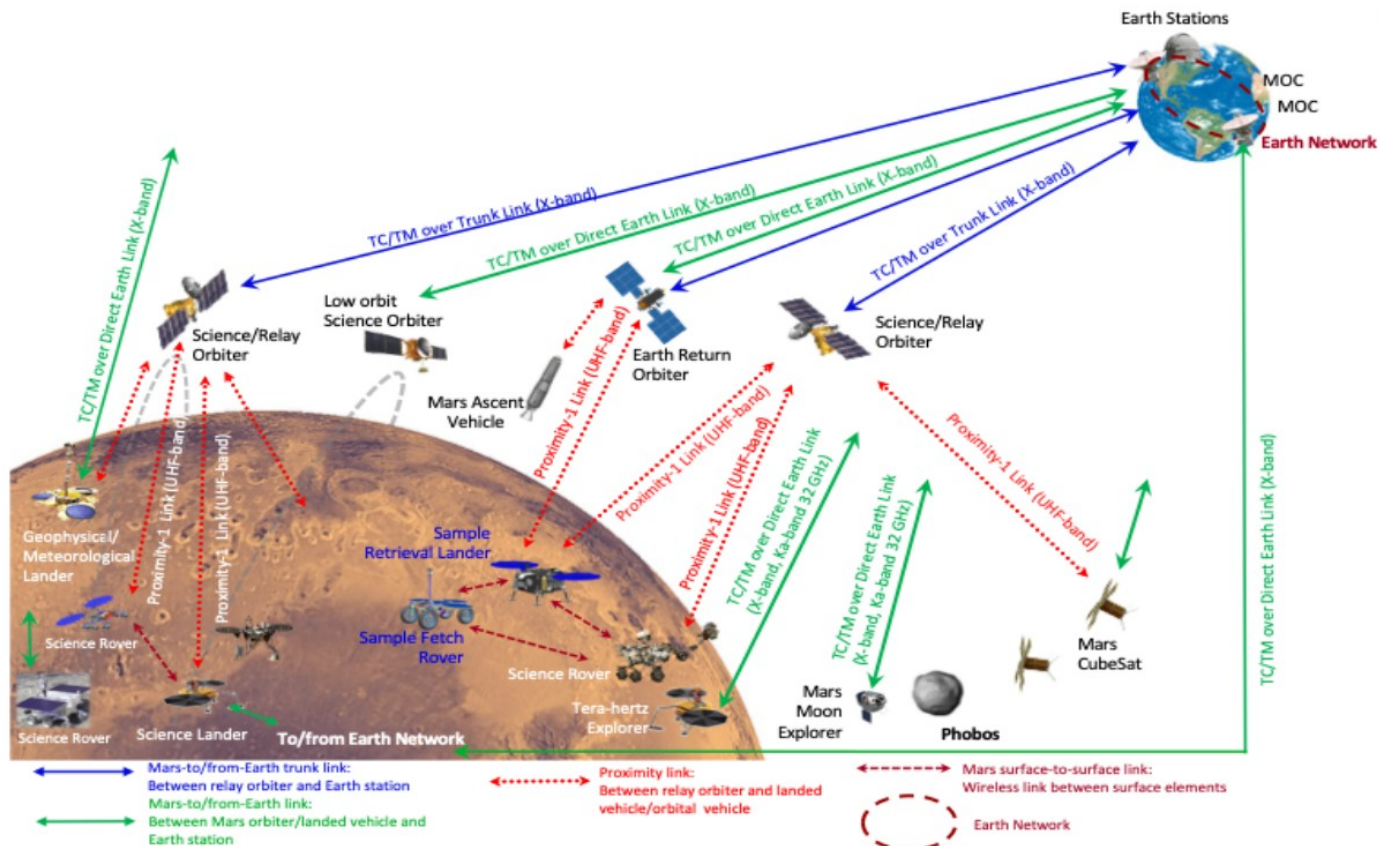
Overview of Data Rates and Volumes

- Increase of expected data volume by x 1000 within 30 years
- **Relay capabilities** as a **primary function** of orbiters is required
- Migration from X-band to **Ka-band** first for downlink and in the long-term mission era for uplink
- **DTN protocols** probably be used in the **mid-term mission era**



Future Near-Term Mission Era

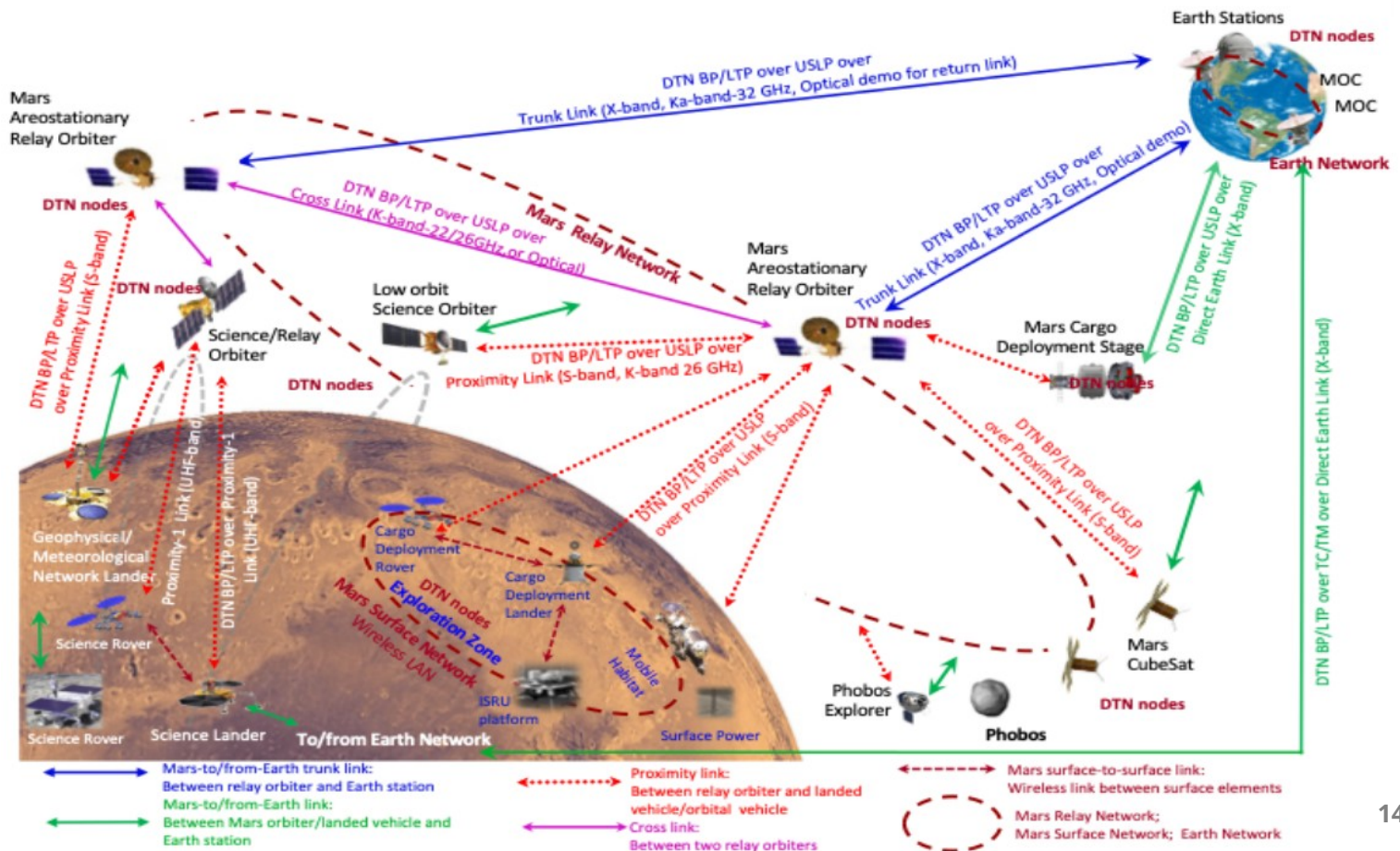
- Several sample return missions, Mars moon exploration missions and several Mars orbiters are planned
- Obviously, in near-term missions the fundamental network architecture will not change
- Several key characteristics will improve and are optimized resulting in a more suitable fundament for a real Mars network



From [2], page 74, figure 5.2-2

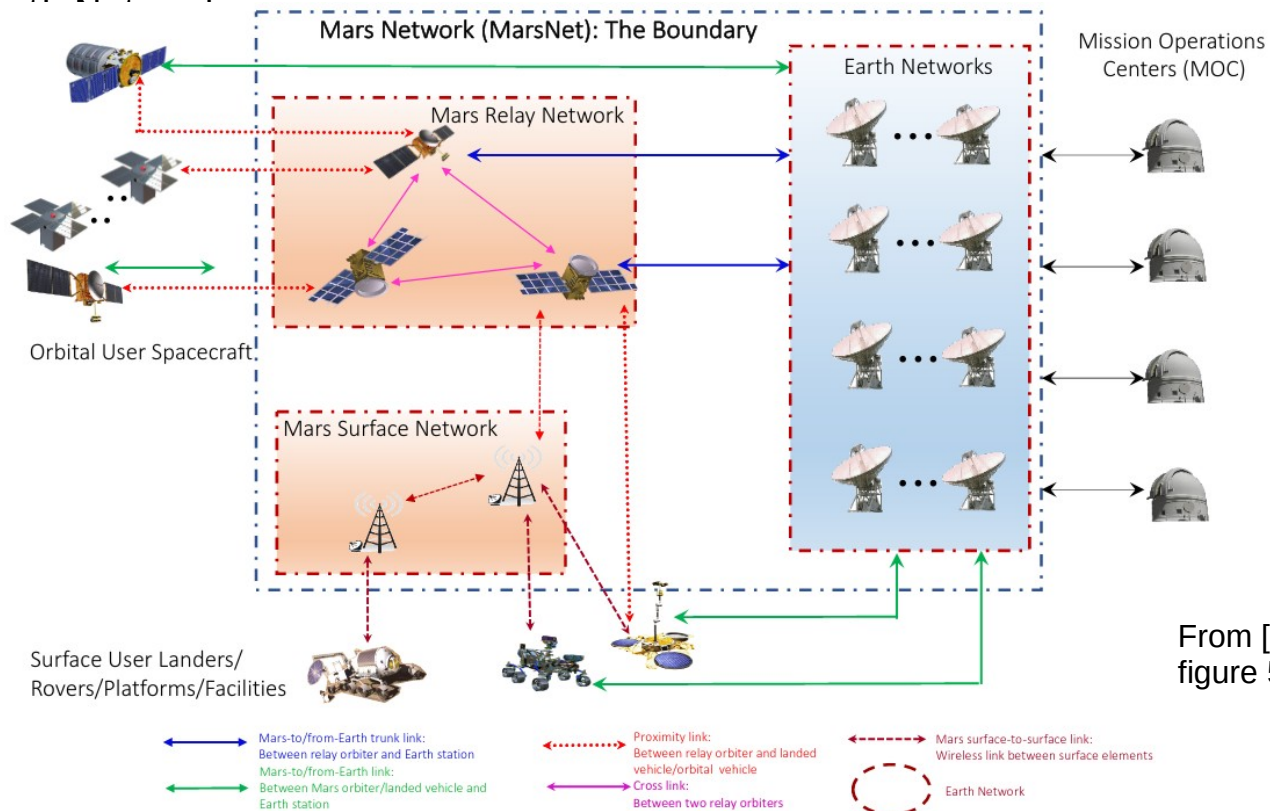
Future Medium-Term Mission Era

- Ka-band capability for direct-to-earth communication will be operational plus optical downlink demonstrations will be done
- Dedicated Mars relay satellites will be deployed in higher orbits (with relaying as primary function, two relay satellites are currently under consideration)
- Early version of an end-to-end MarsNetwork (MarsNet) based on the DTN architecture will be available
-



Mars Network (MarsNet)

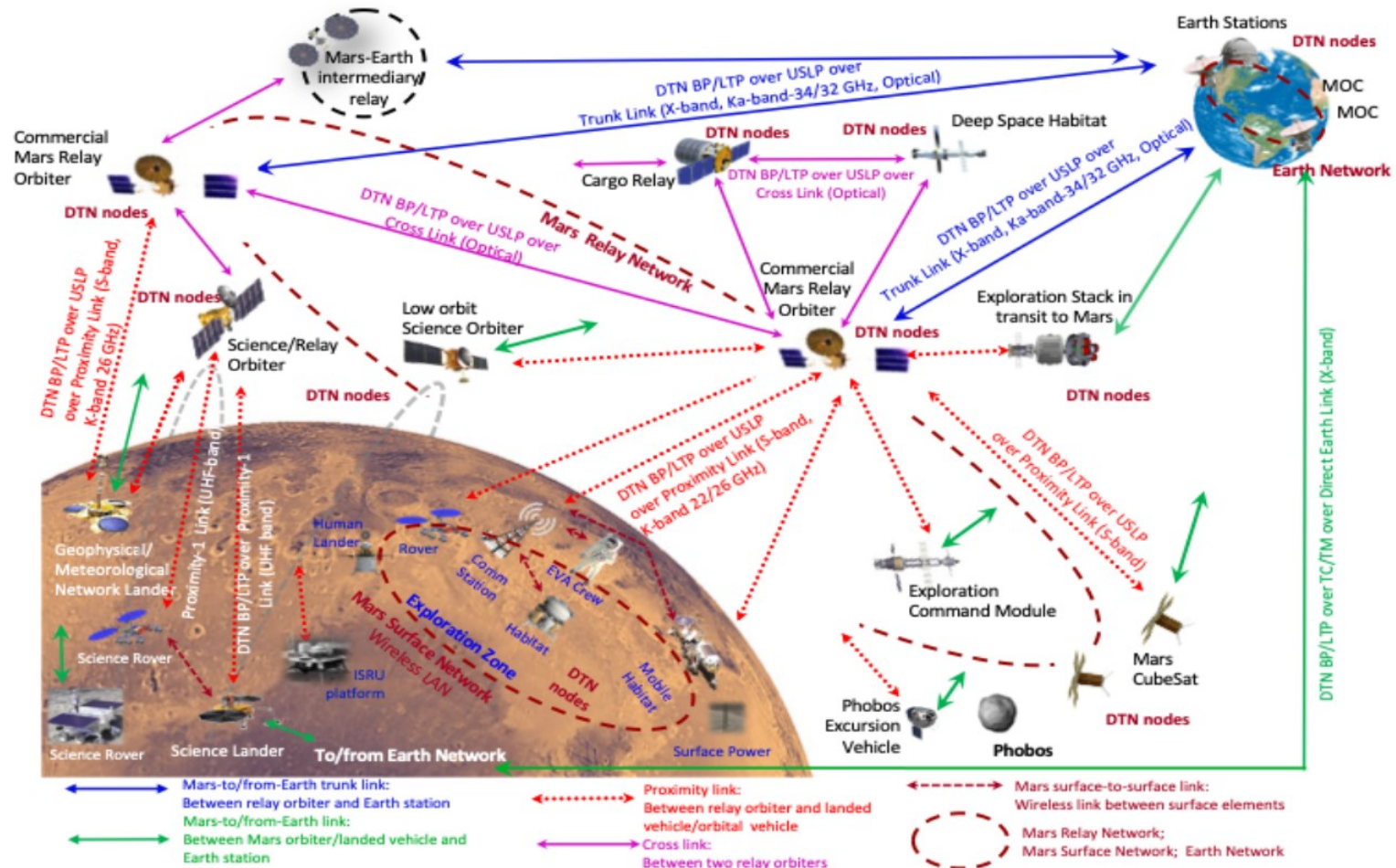
- Establishment of Mars relay network plus use of surface-to-surface links for the interface between ExoMars Kazachok lander and Rosalind Franklin rover and that between MSR's Sample return lander and Sample fetch rover form fundament of the MarsNet (taken from p. 75 of [2])
- DTN protocols bring together the three network segments:
 - Mars Surface Network
 - Mars Relay Network
 - Earth Networks



From [2], page 76
figure 5.2-3

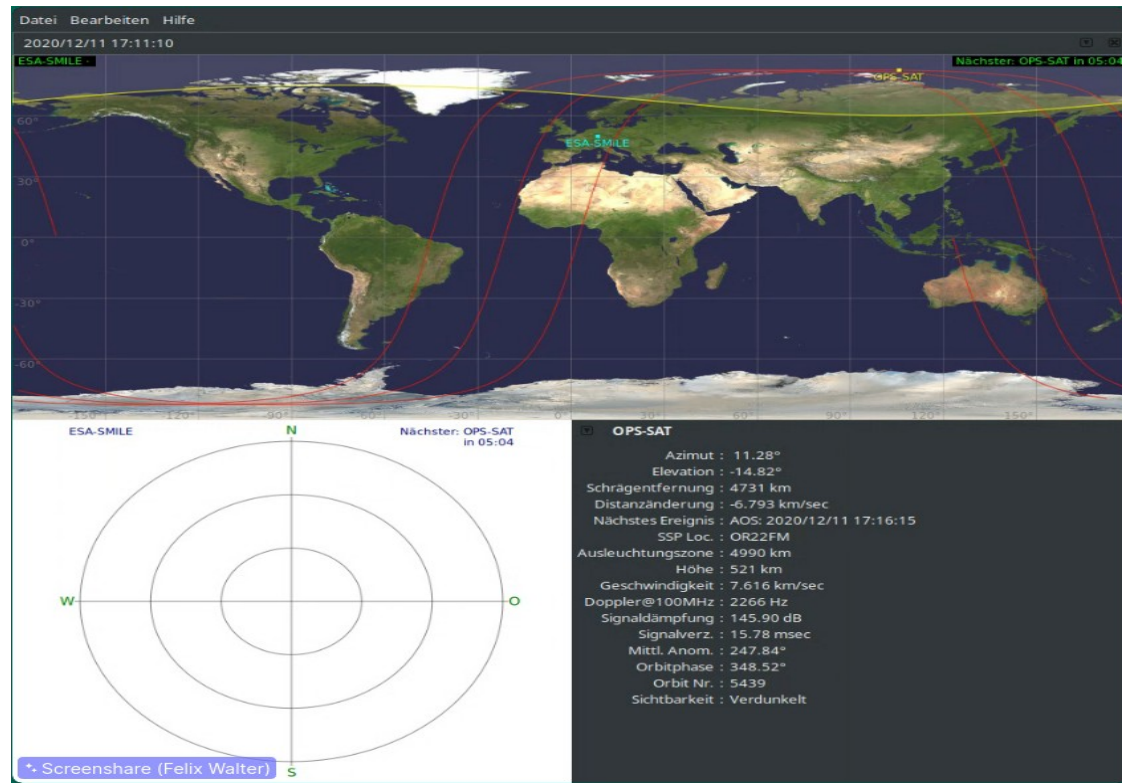
Future Long-Term Missions

- Partially an extension of the network established for mid-term missions, e.g.:
 - Crosslinks between relay satellites
 - Mars surface network may be extended by WiFi/3GPP/LTE networks for human missions
 - Full fledged end-to-end DTN between earth and Mars, all network assets will be treated

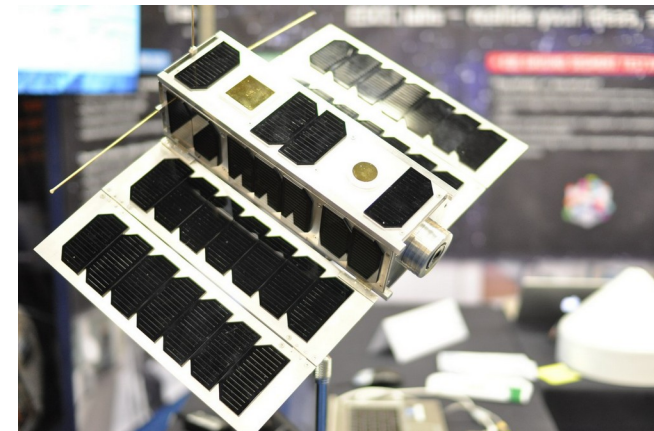


ESA OPS-SAT

- 3U CubeSat launched in December 2019
- Leverages Critical Link MityARM 5CSX with Dual-core 800 MHz ARM Cortex-A9 processors operating Linux
- Offers TCP interface to communication subsystem → SPP is transferred via TCP by the on-board μ D3TN instance
- Access to space segment via ESA SMILE Lab leveraging a reverse SSH port forwarding setup



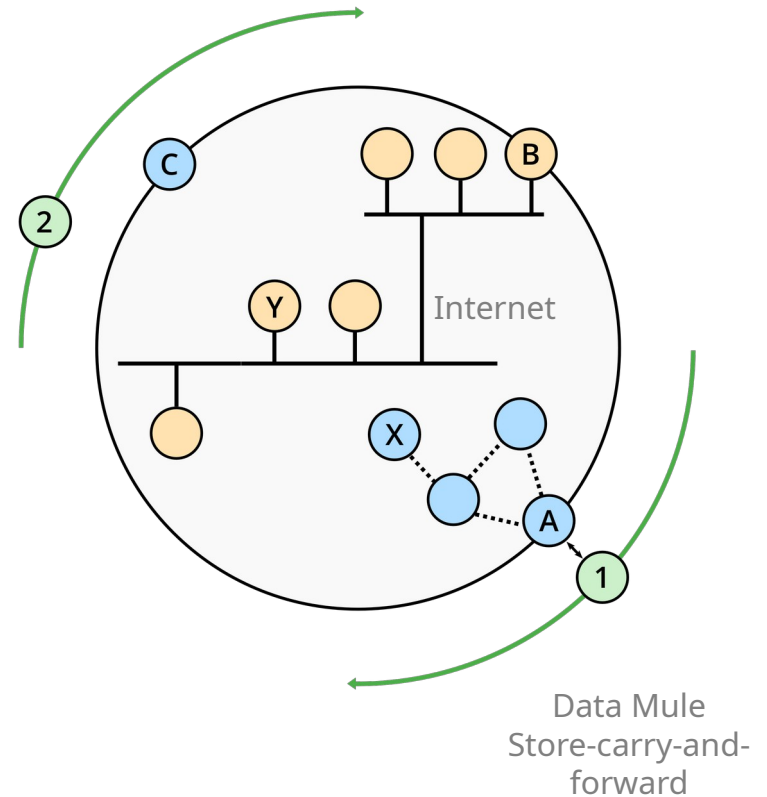
ESA SMILE-Ground Station
Image: ESA



OPS-SAT Model
Image: ESA

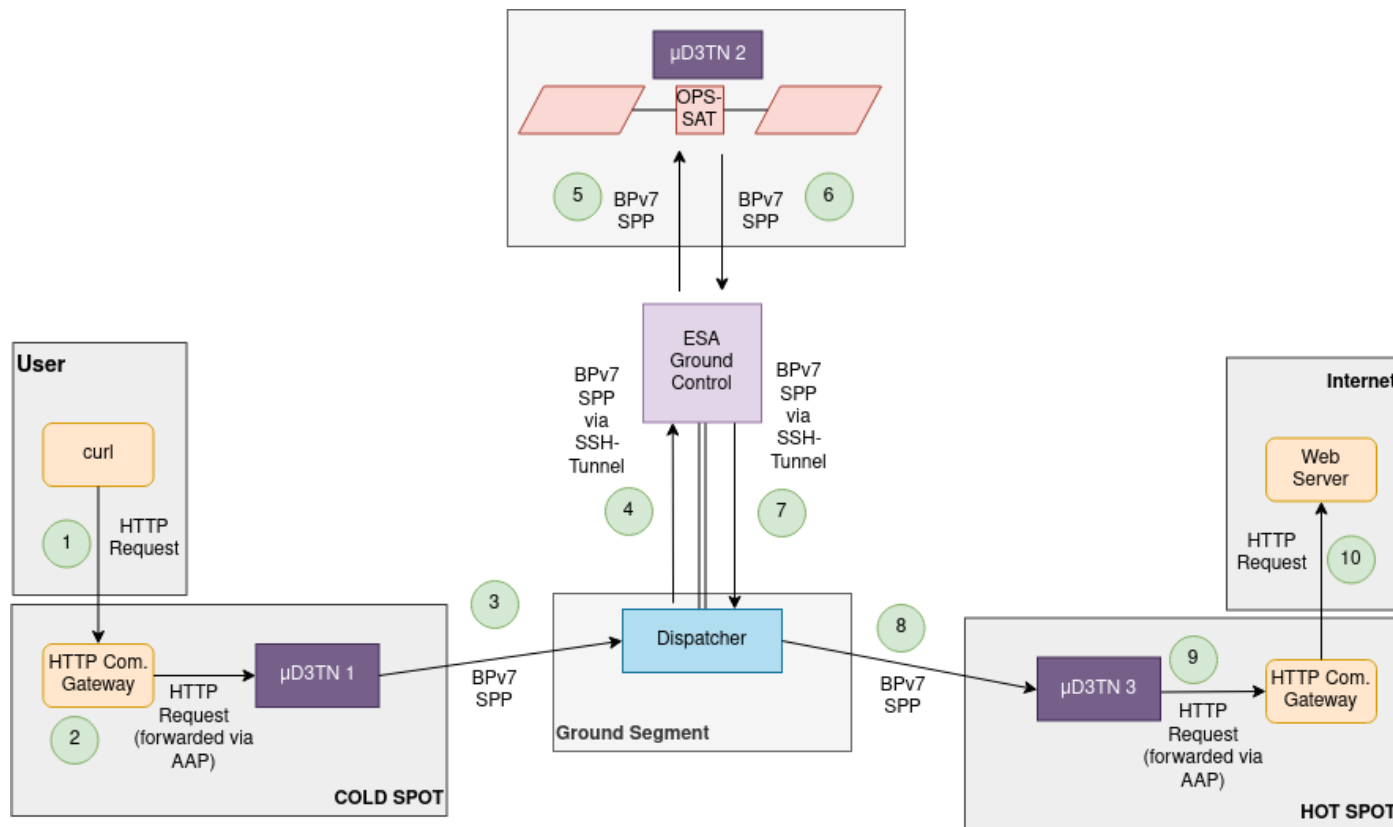
Background: Ring Road Networks

- Ring Road Network (**RRN**) concept
 - **Data Mule**: LEO micro- or nano-satellite
 - **Cold Spot**: no Internet access
 - **Hot Spot**: gateway to the Internet
- Ideal for advancing/testing DTN protocols
 - **Viable** and **scalable**
 - Concrete value for **plenty of use cases**
 - Remote places
 - Other planets



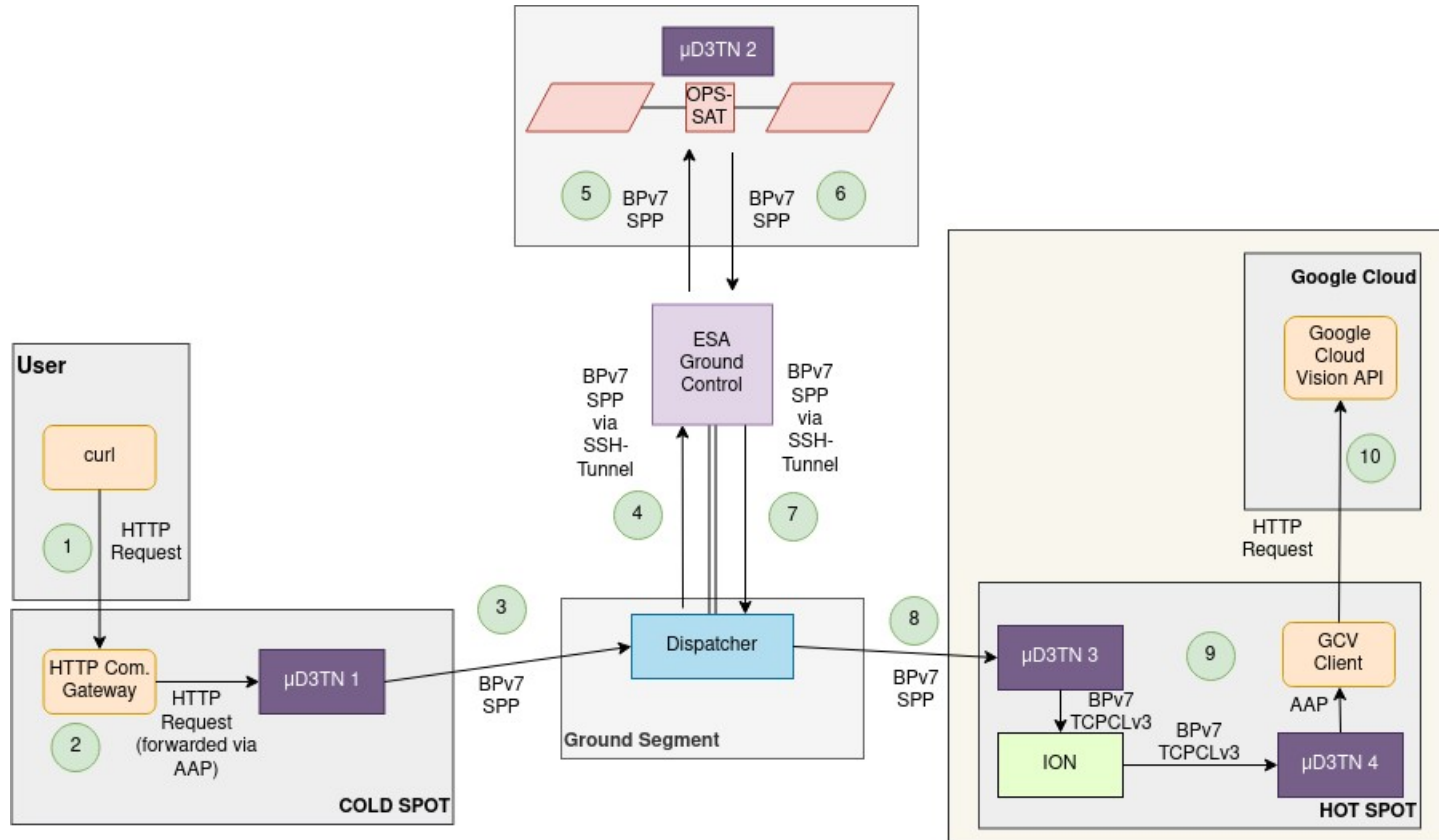
Experiment 1: Access to Web Server

- HTTP request/response are transmitted via Ring Road Network between a remote site and the Internet
- First test of Bundle Protocol version 7 in space
- Experiment leveraged μ D3TN on ground as well as on the satellite



Exp. 2: Interoperability & Cloud Interaction

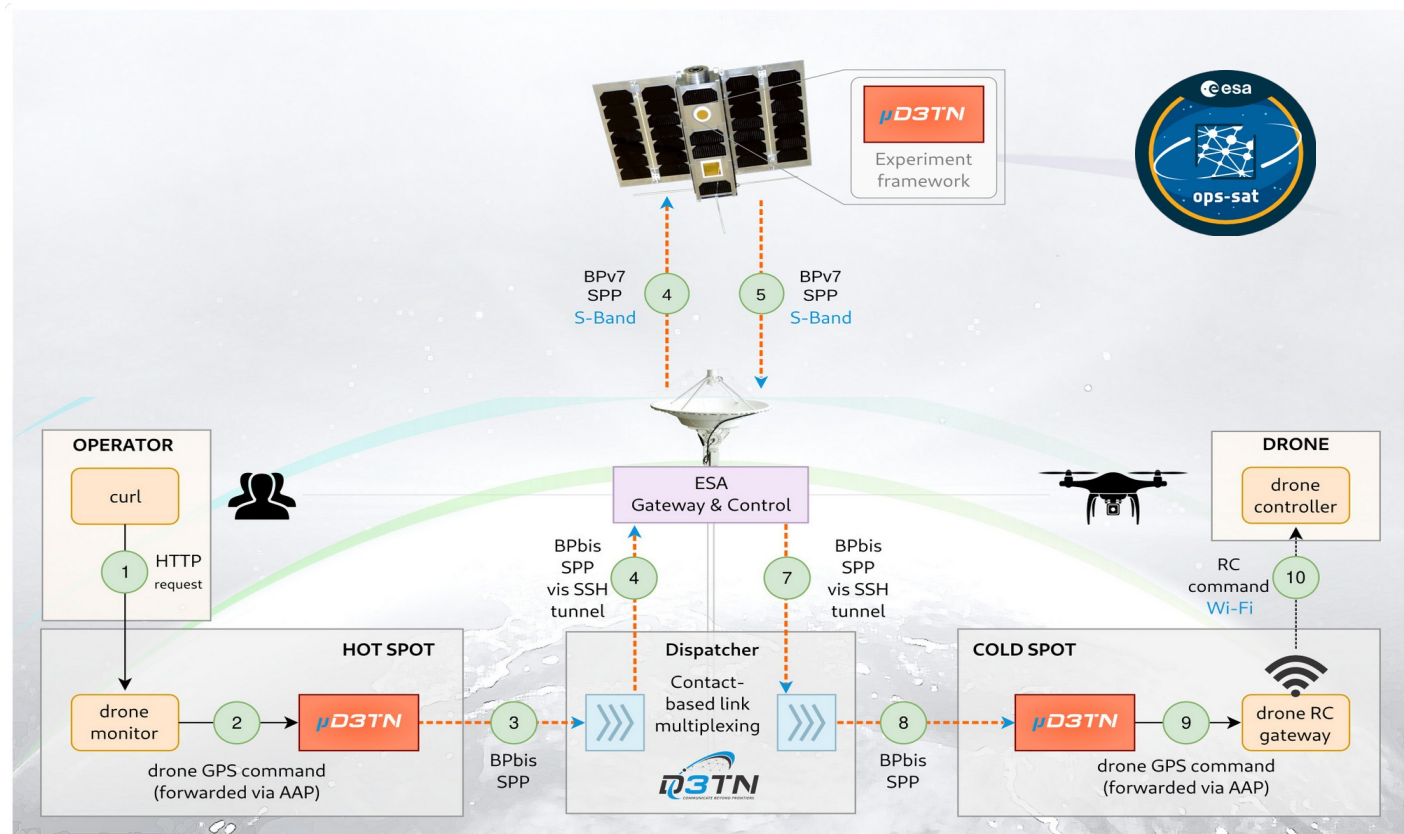
- Experiment has been conducted together with Spatiam Corporation
- Base version shown on the previous slide has been extended to
 - demonstrate compatibility of the BPv7 implementation with ION (DTN protocol implementation developed by NASA)
 - Demonstrate access from a Cold Spot to Cloud services





Experiment 3: REDMARS Experiment

- Goal: Remote UAV control via heterogenous DTN
- **REDMARS** project funded by the Federal Ministry of Education and Research (FKZ16KIS1356) explores options for transferring RINA concepts to the domain of Delay- and Disruption-tolerant Networking

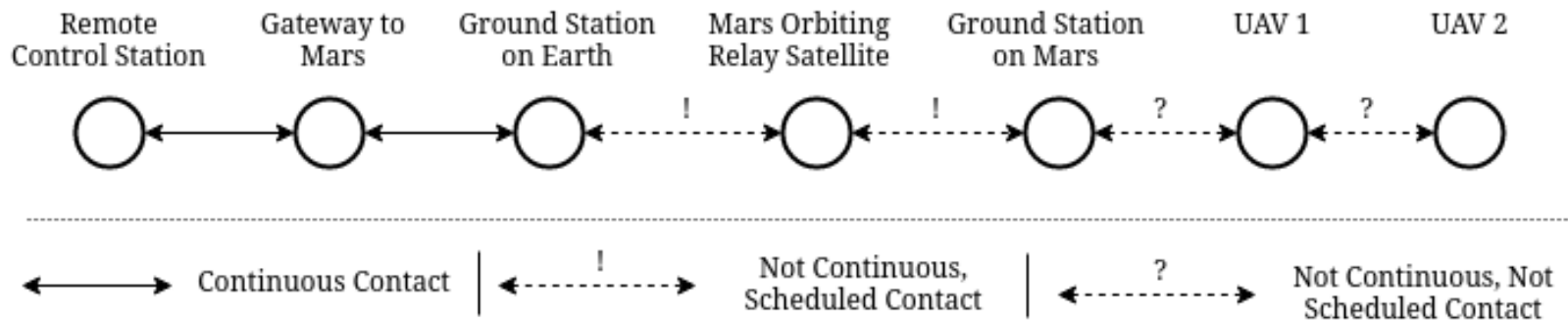


REDMARS Field-test



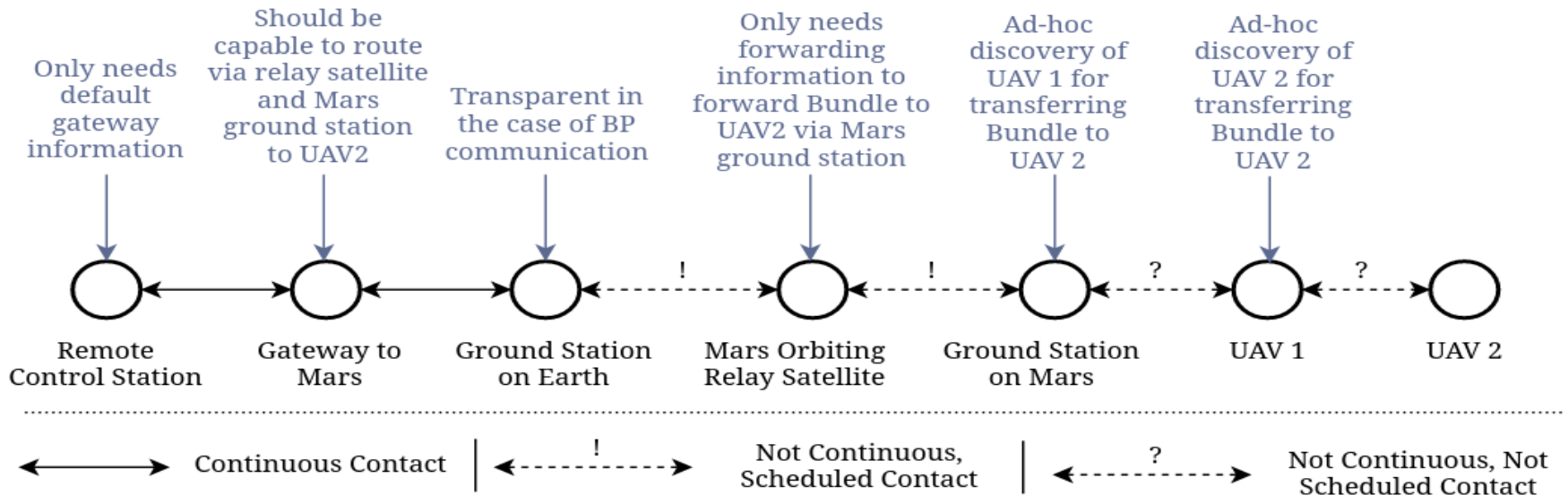
Link Characteristics

- Scenario includes communication links with different characteristics:
 - Continuous links
 - Not continuous links with deterministic behavior
 - Not continuous links without deterministic behavior



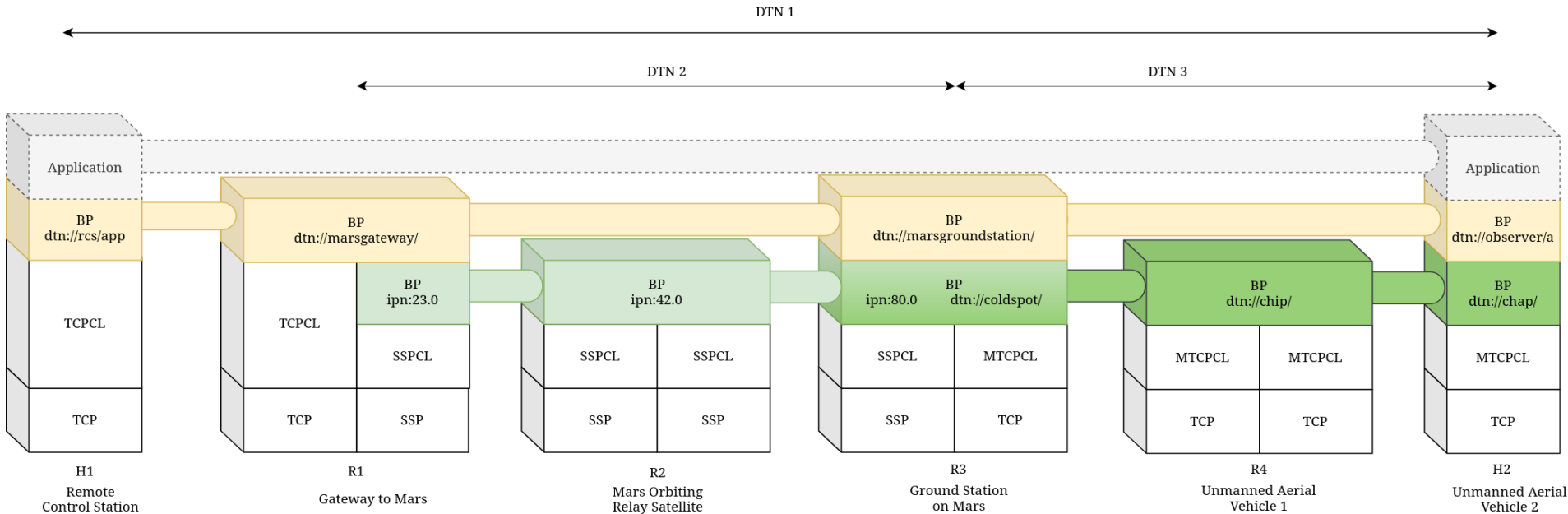
Requirements for Dispatching / Routing

- Requirements for dispatching / routing are very heterogenous, e.g. for communication from the remote control station to UAV 2, the following situation is given:



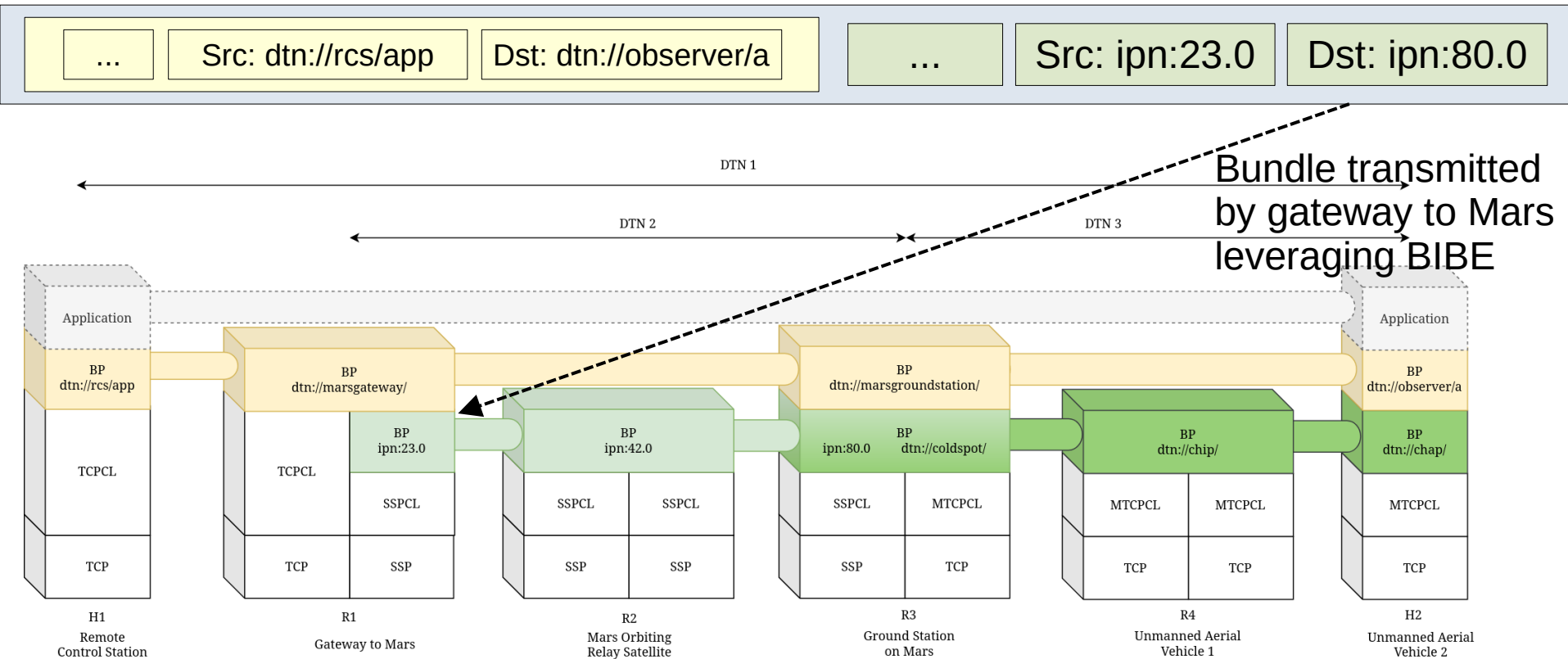
Decomposing the DTN

- There is no single DTN layer but at least two layers of specific scope
- On each layer and node only information should be available required for fulfilling the forwarding function of the node within the layer's scope



Decomposing the DTN

- Routing and forwarding in overlay DTN is decoupled from routing and forwarding in the underlay DTNs
- Underlay DTNs are transparent for overlay DTN due to the usage of Bundle-In-Bundle-Encapsulation (BIBE)
- An overlay BP implementation leverages an underlay BP implementation just as any other CLA



References

- [1] CCSDS Report Concerning Space Data System Standards, "Overview of Space Communications Protocols", Green Book, Information Report, CCSDS 130.0-G-4, April 2023
- [2] Report of the Interagency Operations Advisory Group, Mars and Beyond Communications Architecture Working Group, "Volume 1. The Future Mars Communications Architecture", Final Version, 22 February 2022
- [3] CCSDS Report Concerning Space Data System Standards, "Overview of the Unified Space Data Link Protocol", Green Book, Informational Report, CCSDS 700.1-G-1 June 2020